



A Foundation-Model T-cell Dysfunction Screen: Translational Candidates, Then a Full Audit

Four checkpoint/persistence hits, tissue-validated -- and what a full concordance, ambient, and donor-level audit does and does not change about them.

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Slides, report and build package: github.com/Kays3/talks

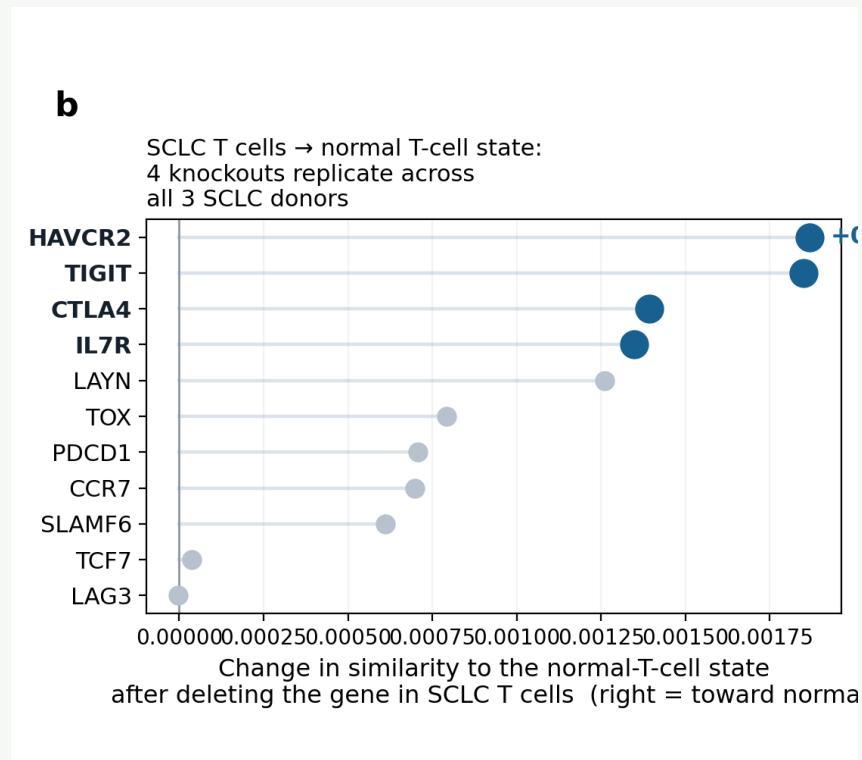
A perturbation screen can look statistically overwhelming and still be wrong

Foundation models fine-tuned on single-cell transcriptomes can be probed for candidate disease-state drivers: delete or overexpress a gene's simulated expression in a held-out cell, and read the shift in a fine-tuned classifier's predicted state. A dose-responsive driver should move the model toward its target when overexpressed and away from it when deleted.

A classifier trained on real tissue also learns whatever else correlates with its labels -- ambient RNA from other cell types, detection-rate differences, cohort composition, tissue of origin. None of that is cell-intrinsic biology, and a screen does not tell you on its own which one it found.

This talk presents a set of translational candidates identified this way, then applies the audit a screen like this needs: a concordance test, a screen-level ambient-contamination check, and a donor-level reanalysis of a cross-cohort ordering claim.

Four replicated checkpoint/persistence hits, tissue-validated



A donor-disjoint SCLC/LUAD/normal T-cell classifier supported an applied in-silico perturbation screen for candidate CAR-T/ICI-relevant drivers. Four hits replicated across the screen: **TIGIT, TIM-3 (HAVCR2), CTLA-4, IL7R.**

Independent Visium spatial validation (GSE263196, 5 SCLC specimens, 15,632 in-tissue spots) supported a tissue-validated antigen-presentation programme.

Presented at JSDP, 2026-08-17. No caveat attached here -- none of these four hits is implicated in anything that follows.

One test was already flagged as pending; one new screen needed the same scrutiny

The poster's own hedge

"The model implies Normal < SCLC < LUAD on the checkpoint axis. This conflicts with the clinical tumour-level picture of SCLC as cold and ICI-resistant, so it **remains an open question rather than a reconciled conclusion.**"

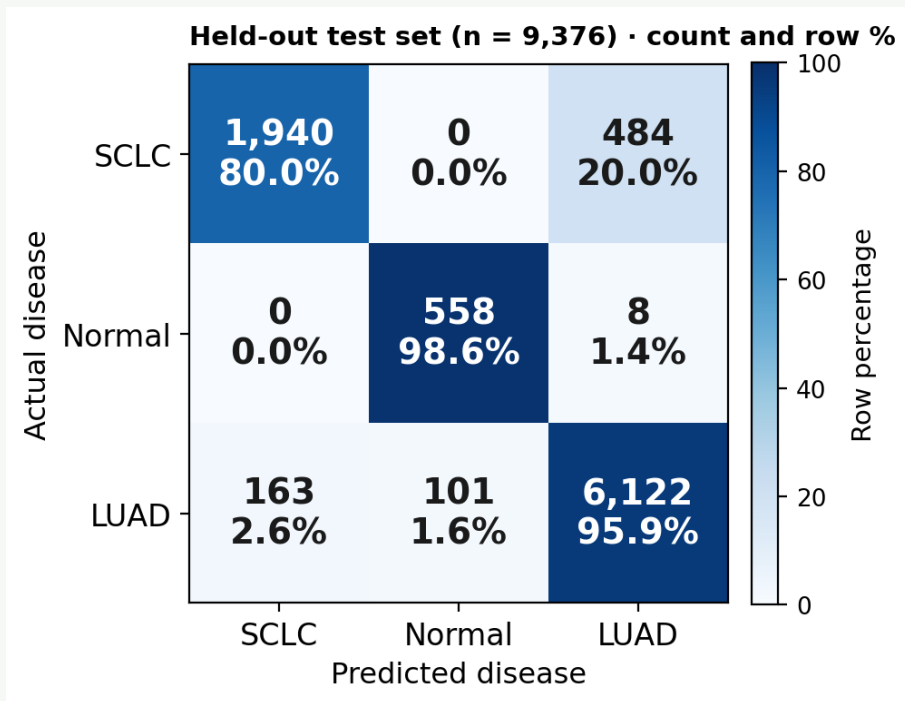
The talk's own QC slide, "TESTED SINCE THE POSTER"

"The states are not on a line, so there is no ordering to contradict... Test 1 of 4 — sign directions, not centroid geometry. **Pending T2 and T3.**" The donor-level test was named as open before this work began.

What this talk adds

A later, expanded screen surfaced two new high-FDR candidates needing the same concordance scrutiny the original four received. This talk **completes the pending test** and applies that scrutiny — it does not reopen a settled claim.

46,140 T cells, 42 donors, one fine-tuned classifier



Held-out test performance

91.9% accuracy, macro F1 0.903, donor-disjoint split. Geneformer V2 104M fine-tune.

Normal-class test performance rests on 1 held-out donor (566 cells)

A third of this macro-F1. Already flagged in the repo's own methods

(perturbation_workflow/METHODS.md, lines 57-67 the split table, line 112: "Normal-class metrics are flagged explicitly given 1 test donor").

19 SCLC + 22 LUAD + 4 normal = 45, but 42 unique donors: three donor_ids (RU675, RU682, RU684) each carry two biospecimens from the same clinical patient, one LUAD tumour and one normal-tissue sample (METHODS.md, lines 33-39).

Delete or overexpress, in held-out cells, across six comparisons

Each gene is deleted (its token removed from a cell's rank-ordered input) or overexpressed (promoted to the front of the rank) in held-out cells, and the shift in the classifier's predicted disease-state probability is read for both operations.

Six source→goal disease-state comparisons: SCLC→LUAD, SCLC→normal, LUAD→SCLC, LUAD→normal, normal→SCLC, normal→LUAD.

A dose-responsive driver should move the model toward its target under overexpression and away from it under deletion — the two operations are read together, never one alone.

A real driver must move opposite ways under deletion and overexpression

The concordant **criterion**, used throughout this repo's perturbation work:

```
delete_fdr < 0.05 AND overexpress_fdr < 0.05 AND delete_shift × overexpress_shift < 0 AND delete_n ≥ 25
```

Significance in one arm is not enough. A gene must clear both arms and reverse sign between them — the signature a dose-responsive driver actually produces. A gene can be significant in both arms and still fail this test if deletion and overexpression push the classifier the **same** direction — the model reacting to the gene's presence at all, not to a dose-responsive relationship with the labelled state.

Two further checks, applied after the screen, not instead of it

Ambient-risk audit

Every top-effect gene is checked against a curated ambient-contamination panel and a curated T-cell-intrinsic anchor panel. The flag threshold is the 25th percentile of the full ambient-anchor population's own risk scores — a within-panel, not an absolute, cutoff.

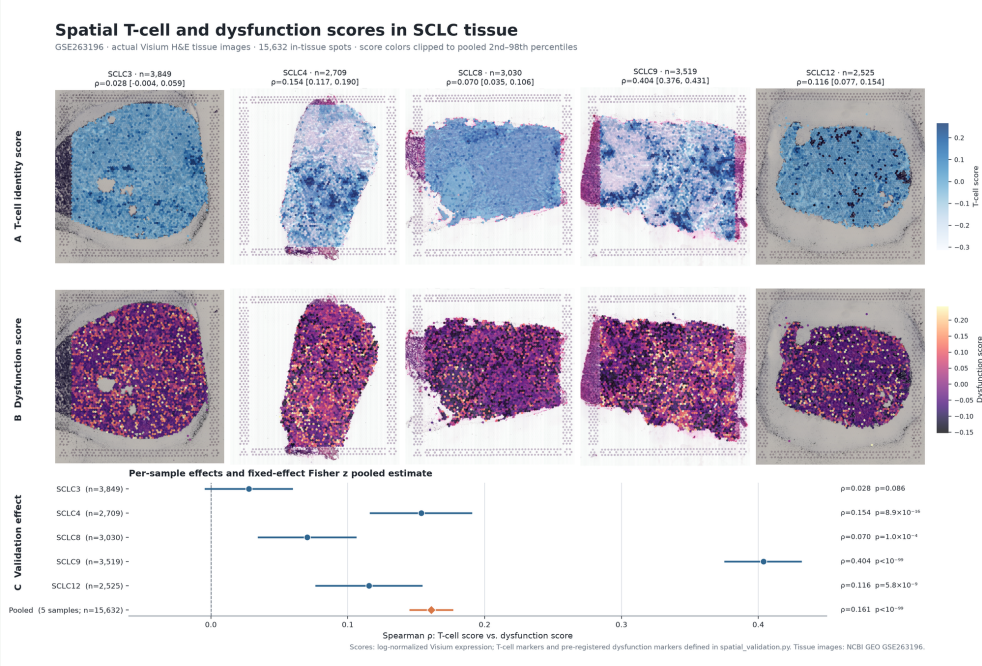
Donor-weighted reanalysis

Any cross-cohort comparison is recomputed donor-level (mean within donor first, then an unweighted mean of donor means) alongside the original cell-weighted statistic, to separate a real group difference from one donor's cell count dominating the mean.

Why both, not either

A significance test alone cannot distinguish a cell-intrinsic effect from an ambient or cell-count artifact. Both audits are applied to every candidate discussed in Results, not selectively to ones that already look questionable.

Four hits replicate and validate independently in tissue



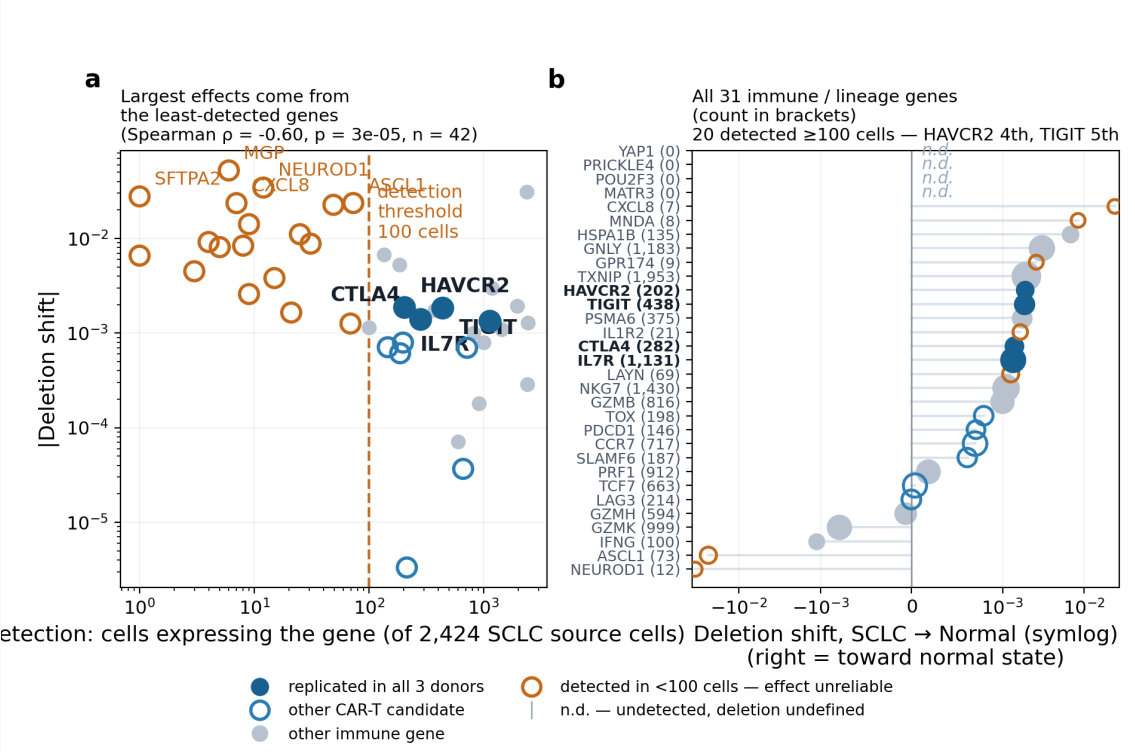
TIGIT, TIM-3 (HAVCR2), CTLA-4, IL7R

Four replicated checkpoint/persistence edits from the applied screen (screen_b.png, Introduction). Replication here means: deletion FDR < 0.05 and the same deletion sign in all 3 SCLC donors.

GSE263196 Visium, 5 SCLC specimens, 15,632 in-tissue spots: antigen-presentation programme $\rho = 0.361$, 7.95 sigma above the null.

The concordance half of the original call was made by the pre-2026-09-22 panel runner, which overexpressed into all 2,424 SCLC cells while deleting only in the 202–1,131 cells expressing each gene; it has not been rerun with paired cell sets. Unaffected by S100, ambient or T6 below; the spatial number is untouched. A qualifier, not a retraction.

Sparse genes fake big effects, and the poster's own ordering fails its own test

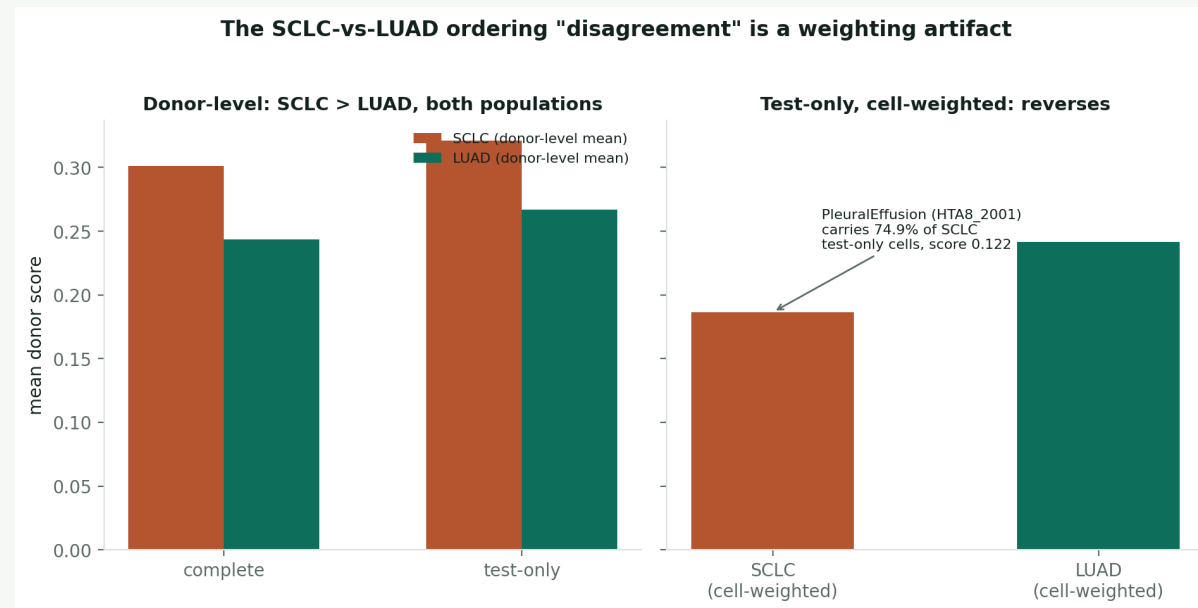


Detection vs. effect size: Spearman $\rho = -0.60$ ($p = 2.7e-5$, $n = 42$) across the full 50-gene panel.

“TESTED SINCE THE POSTER” (talk, slide 7): “the states are not on a line, so there is no ordering to contradict... Working reading: a triangle — $SCLC \leq \text{Normal} < \text{LUAD}$... Test 1 of 4 — sign directions, not centroid geometry. **Pending T2 and T3.**”

This is the original team's own self-correction, dated before this work began — not a finding of ours.

The open question is now closed as null



Donor-level: no reversal

Complete population: SCLC 0.301 > LUAD 0.244. Test-only: SCLC 0.321 > LUAD 0.267. Two-sided **Monte Carlo** test, 100,000 replicates: $p = 0.216$, 19 vs. 22 donors. **Not significant.**

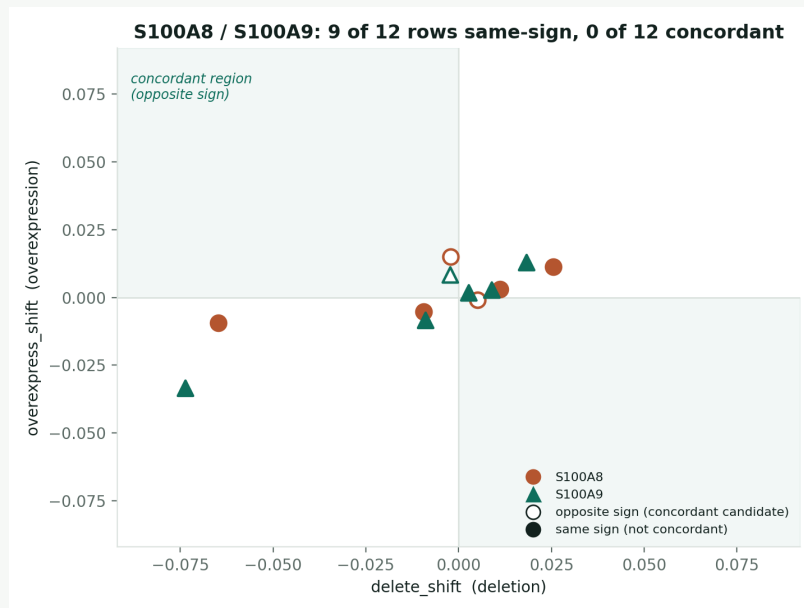
PleuralEffusion: one genuine donor

Carries 74.9% of test-only SCLC cells.

HTAN_Participant_ID confirms exactly one participant (HTA8_2001), not several merged (METHODS.md, lines 29-32).

A separate, **exact-enumeration** test on the 3-vs-4-donor test-only slice: only 35 label assignments exist; the smallest attainable two-sided p is 0.0571 (2/35) — that design cannot reach significance regardless of the true effect.

S100A8 and S100A9: an overwhelming FDR, and no concordance at all



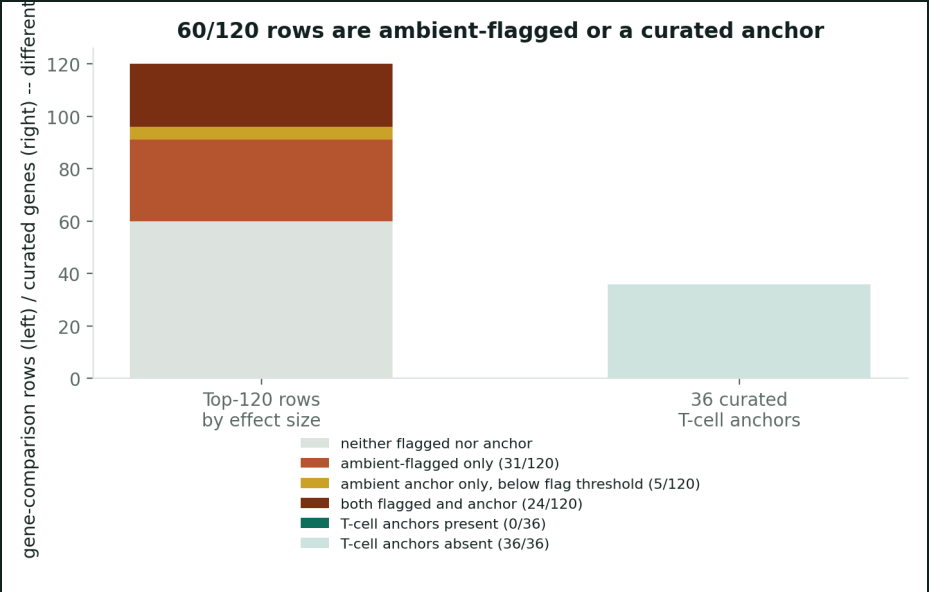
9 of 12 rows same-sign; 0 of 6 comparisons concordant

Example: S100A8, LUAD→SCLC deletion — FDR = 1.45e-224, N = 713 detections, shift = +0.0255.
The FDR is a certainty claim, not a size claim — it says nothing about direction. First appearance of these two genes in this deck: caveated from the start, not a later correction.

What the LUAD-normal and SCLC-normal rows can and cannot lean on

Of 12 rows, 8 touch the normal class (4 comparisons). For luad↔normal, 3 of 4 normal donors are the *same patients* as 3 LUAD donors — real protection against a separate-cohort confound, though only 3 of 4. **No such pairing exists for sclc↔normal.**

Half the top 120 is ambient-flagged or a curated ambient anchor



60/120

rows are ambient-flagged, a curated ambient anchor, or both: 31 flagged-only, 5 anchor-only, 24 both — a real partition of the 120, not two overlapping counts added together

5/120

rows are anchors below the flag threshold — mechanical, not a miss: the threshold is the anchors' own 25th percentile of risk, so some anchors falling under it is built in.

0/36

of the 36 curated T-cell-intrinsic anchor genes (a different denominator: the full curated list, not a row count) appear anywhere in the 120.

One candidate survives every check

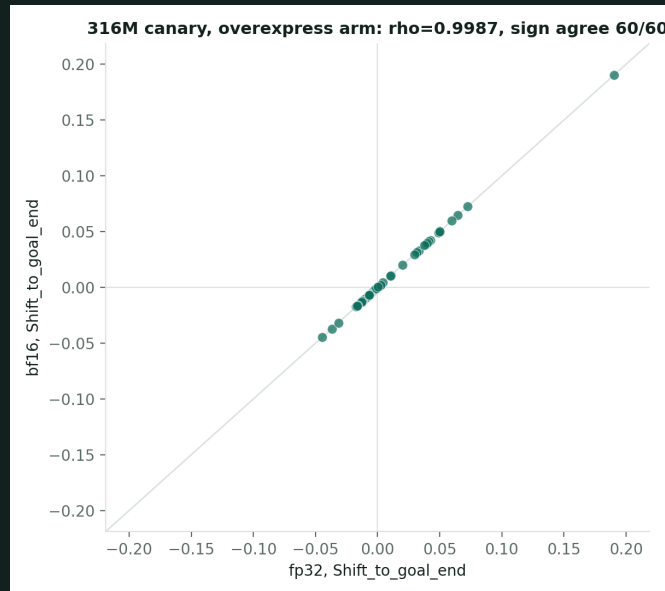
NBEAL1

5 of 5 comparisons concordant — every one it appears in — detected in 75.0% of cells.

Passes the same concordance test that eliminates S100A8/S100A9, at a detection rate well above the ambient range flagged for those two. Does not make it a validated driver on its own — makes it the one candidate this screen has not been able to eliminate.

Not everything restored from the original screen fails, and the one thing that doesn't fail is more informative for having survived a real test.

No: precision replicates cleanly; model size is the axis that moves things



Same model, fp32 vs. bf16 (half precision)

104M panel (300 rows): rho = 0.9998, sign agreement 298/300. 316M canary (60 rows): rho = 0.9987, sign agreement 60/60.

The pre-registered 104M sign-agreement gate is still recorded **FAIL** — a bitwise-deterministic fp32-vs-fp32 rerun made the noise floor zero, not a usable sign threshold; the domain reviewer's ruling was that the frozen gate stands unamended, not that bf16 disagrees with fp32. Both are true at once.

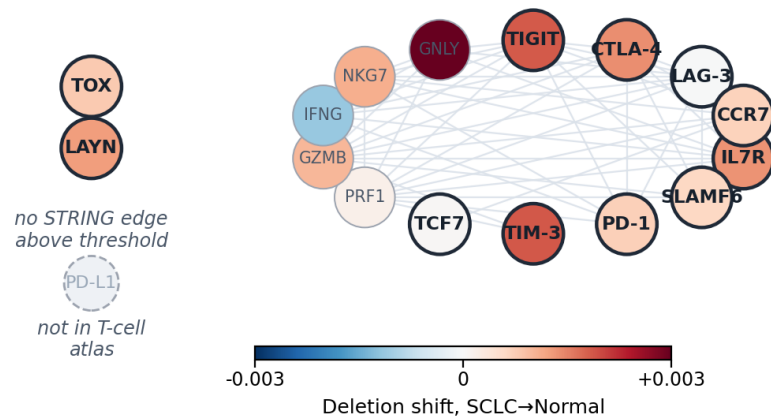
Same model, changed precision vs. changed size

Precision (104M fp32→bf16): rho = 0.9998. Size (104M→316M, both bf16): rho = 0.489. Using (1-rho) as the divergence measure, size moves results ~2,500x (~3.4 orders of magnitude) more than precision. None of this deck's findings are precision artifacts.

The four original hits sit in one connected interaction neighbourhood

C

All 11 candidates on the STRING map;
bold = panel b candidate,
colour = its SCLC→Normal shift



From prior literature, not inferred from these cells

TIGIT: CAR-T support. TIM-3: solid-tumour knockout precedent. CTLA-4: real signal, less CAR-T-specific. IL7R: persistence, not exhaustion. 54 STRING edges among 16 context genes; TOX and LAYN have no edge above threshold; PD-L1 is absent from the T-cell atlas.

No caveat — these four hits are unaffected by anything in this deck's audit.

What this establishes, and what it rules out

Establishes

Four checkpoint/persistence hits (TIGIT, TIM-3, CTLA-4, IL7R) replicate and validate independently in spatial tissue data. A concordance test and a screen-level ambient audit are necessary companions to significance for any further perturbation-based driver claim. One further candidate (NBEAL1) clears every check applied.

Resolves

The checkpoint-axis ordering the original team flagged as open ("pending T2 and T3") is closed: donor-level SCLC-vs-LUAD shows no significant difference in either direction, once a single donor's cell-count share is accounted for.

Rules out

S100A8/S100A9 as T-cell-intrinsic dysfunction drivers, at high confidence, independent of model precision. Model precision (bf16) as an explanation for any finding in this deck.

What this study does not establish

Detection-confound axis is an imperfect proxy

Ambient risk is scored partly against the classifier being audited — evidence of resemblance to known contaminants, not a direct contamination measurement.

Single cohort; curated lists

One HTAN/CELLxGENE cohort; the ambient-anchor and T-cell-anchor gene lists are curated, not ground truth. A different cohort or anchor set could shift specific gene calls.

Thin normal class, both counts

1 held-out test donor AND 4 cohort donors overall — already a named limitation in the repo's own methods (METHODS.md, lines 24-25), carried forward here, not newly flagged.

Tissue-of-origin skew, per contrast

SCLC: 7/19 (37%) lung, 12/19 metastatic.
LUAD: 16/22 (73%) lung. Normal: 4/4 (100%) lung. Skews
sclc_to_normal/normal_to_sclc, not
luad_to_normal/normal_to_luad (which
benefits from the 3-of-4 shared-patient
pairing above).

Open questions: the original talk's CRISPR TIGIT/HAVCR2 knockout proposal; does NBEAL1 replicate independently; the talk's own still-open question — “if SCLC T cells are not exhausted, what are they, and why does ICI still fail?” A clean negative, honestly obtained, is a contribution.

